

Lokesh Reddy Sodanapalli

Embedded Test & Automation Engineer | System Integration & Validation

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github.com/Lokisdp lokisdp.github.io/lokesh_portfolio

Languages: English - C1 (IELTS) | German - B1

PROFESSIONAL EXPERIENCE

01/2026 – Present
Dresden

JOYNEXT GmbH
System Test & Automation Engineer | Working Student

- Derived 48 tests/procedures from Polarion requirements; linked results and GitLab CI execution.
- Built Python/Robot automation for ~80% of recurring tests; developed 8+ reusable bench libraries(CAN, TCP/IP,DoIP, Serial, Rest API, SSH).
- Verified RTCU(ECU) hardware/software, power and lifecycle states; analysed faults and verified fixes.
- Built embedded C firmware on ESP32-C5 for stepper-driven ECU wake-up; debugged hardware/firmware.

12/2024 – 07/2025
Berlin

MBition GmbH
Embedded Software Test Engineer | Working Student

- Derived connectivity tests and procedures; automated regression with Python and Robot Framework.
- Verified Linux and QNX ECUs through functional, integration and regression testing.
- Configured test benches and flashed ECUs; supported Jenkins builds and software integration.
- Investigated DLT logs, system traces and network data; documented faults and verified fixes.

05/2024 – 10/2024
Ulm

Continental ADC GmbH (AUMOVIO)
System Integration & Test Engineering Intern

- V-model: analysed requirements, derived cases/procedures and recorded evidence.
- Commissioned 1:1 vehicle-scale HIL: five radars, HPC, Ethernet switch, relays and Arduino control.
- Tested real ECUs/benches with pytest; PyQt5 flashing eliminated 90% of manual flashing steps.
- Configured sensors and measurement hardware; debugged ECU communication using CANoe/Wireshark; evidenced a ~95% CPU-load anomaly.

06/2021 – 08/2022
Bengaluru, India

Tata Consultancy Services Ltd.
Assistant System Engineer – Validation & Test

- Derived test plans, specifications and manual/automated ECU tests from system requirements.
- Executed functional, non-functional, system, regression and release-validation tests.
- Analysed CAN communication in CANalyzer; documented defects and verification evidence in Jira/X-Ray.
- Collaborated with developers and product owners in Agile/Scrum on retesting and fix verification.

EDUCATION

10/2022 – 09/2026
Chemnitz

Technische Universität Chemnitz
Master of Science - Embedded Systems

07/2017 – 06/2021
Chennai, India

Vellore Institute of Technology
Bachelor of Technology (B.Tech.) – Electrical and Electronics Engineering

TECHNICAL SKILLS

Verification & Test: Requirements-based testing, test case and procedure derivation, test planning, test environment setup, HIL test benches, hardware-related software testing, verification documentation, requirements traceability (Polarion), regression testing, root-cause analysis

Programming & Automation: Python, C, Embedded C, Robot Framework, PyTest, PyQt5, GitLab CI, Jenkins, Git

Microcontrollers & Interfaces: STM32, ESP32-C5, Arduino, GPIO/I2C/SPI/UART, CAN, CAN-FD, LIN, Automotive Ethernet, TCP/IP, XCP

Debugging & Measurement: Oscilloscopes, signal generators, multimeters, SCPI power supplies, CANoe, Wireshark, DLT Viewer

Standards: ISTQB CTFL, ISO 26262 , ISO/SAE 21434 awareness

ACADEMIC PROJECT

01/2026 – 09/2026 **Validation of Radar Based Ego-Motion Algorithms Using Synthetic Radar Data**
Master Thesis

- Developed a Python-based SiL validation framework; used PyTest to write and execute automated test cases.
- Analysed system performance using NumPy, SciPy, Kalman filtering, and RMSE/bias/drift KPIs; applied Codex and ChatGPT for test code generation and debugging.

10/2023 – 04/2024 **AC Analysis and Impedance Measurement on STM32**
Master Project

- Built an embedded impedance measurement system on STM32F3; implemented Goertzel, one-point DFT, and one-point DTFT in embedded C.
- Operated oscilloscopes, signal generators, and multimeters; used MATLAB for signal generation and analysis; validated via SPI and UART.

09/2026 – 09/2026 **ECU IMU Wake-Up Test Automation**
Professional Project

- Built an ESP32-C5/stepper-motor setup to rotate an ECU and trigger IMU wake-up.
- Integrated embedded C motor control with Python/Robot Framework via serial commands.

CERTIFICATIONS & TRAINING

ISTQB Certified Tester – Foundation Level | Bengaluru, India

Automotive Ethernet Basics – Protocols & Network Stack (PHY–MAC–TCP/IP–SOME/IP–DoIP)
Ethernet, IPv4/IPv6, TCP/UDP, ARP, ICMP, switching, routing, SOME/IP and DoIP

A AN: CANoe Basic
Continental Learning Platform
Measurement Setup, Simulation, CAPL, Diagnostics

Functional Safety Awareness: ISO 26262 Essentials – JOYNEXT Learning Platform

Automotive Cybersecurity: ISO/SAE 21434 – JOYNEXT Learning Platform

Integrated Management System – JOYNEXT Learning Platform
ISO 9001-related quality and process knowledge



Lokesh Reddy Sodanapalli
Chemnitz, 06.10.2026